STATS-HW-3

1.

Outcome <chr>	beta1_dual <dbl>	beta1_multi <dbl>	Bias <dbl>
count_visit_dr	0.3957	0.5935	-0.1978
count_visit_er	-0.0359	0.0023	-0.0381
out_of_pocket_spend	-73.0442	-67.2026	-5.8416
health_score	-0.0016	0.0005	-0.0021
happy	0.0124	0.0051	0.0073

None of the biases are large enough to alter the qualitative conclusions of the OHIE.

```
###1
```{r }
ohie_data <- read.csv("/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw3/OHIE.csv")
y_vars <- c("count_visit_dr", "count_visit_er", "out_of_pocket_spend", "health_score", "happy")
# 空表存储
results_matrix <- matrix(NA, nrow = length(y_vars), ncol = 3)
colnames(results_matrix) <- c("beta1_dual", "beta1_multi", "Bias")
rownames(results_matrix) <- y_vars

# 循环计算每个结果变量的偏差
for (i in 1:length(y_vars)) {
  y <- y_vars[i]
  # 双变量回归：Y ~ treated
  reg_dual <- lm(as.formula(paste(y, "~ treated")), data = ohie_data)
  # 多元回归：Y ~ treated + numhh_list
  reg_multi <- lm(as.formula(paste(y, "~ treated + numhh_list")), data = ohie_data)
  # 提取系数并计算偏差
  results_matrix[i, "beta1_dual"]  <- coef(reg_dual)["treated"]
  results_matrix[i, "beta1_multi"] <- coef(reg_multi)["treated"]
  results_matrix[i, "Bias"]        <- results_matrix[i, "beta1_dual"] - results_matrix[i, "beta1_multi"]
}

# 把数值矩阵和字符列合并数据框
bias_result <- data.frame(
  Outcome = y_vars,
  round(results_matrix, 4)
)

cat("问题1 遗漏变量偏误结果")
print(bias_result)
```
```

2.

Group <chr>	beta1_treated <dbl>	SE_beta1 <dbl>	t_stat <dbl>	p_value <dbl>	Sample_Size <int>
Female (female=1)	0.7926	0.3119	2.5417	0.0111	6916
Male (female=0)	0.3420	0.2836	1.2060	0.2279	5313
Age ≥ 50 (age_group=1)	0.4097	0.3934	1.0416	0.2977	3375
Age < 50 (age_group=0)	0.6440	0.2588	2.4882	0.0129	8855
White (race_white=1)	0.8067	0.2760	2.9226	0.0035	8429
Non-White (race_white=0)	0.1639	0.3303	0.4961	0.6199	3839
Good Baseline Health (health_baseline=1)	0.4180	0.4681	0.8930	0.3719	3314
Poor Baseline Health (health_baseline=0)	0.6450	0.2394	2.6947	0.0071	8915

Magnitude of Treatment Effect:

The largest estimated treatment effect is observed among White individuals ($\beta_1 = 0.8067$), followed by Females ($\beta_1 = 0.7926$) and those with Poor Baseline Health ($\beta_1 = 0.6450$). In contrast, Non-White individuals show the smallest effect ($\beta_1 = 0.1639$), which is less than one-fifth the size of the effect for

White individuals.

Statistical Significance:

At the 5% significance level, the positive effect of insurance on doctor visits is statistically significant for four groups: Females, those under 50 years old, White individuals, and those with Poor Baseline Health. For Males, those aged 50 and over, Non-White individuals, and those with Good Baseline Health, the effect is not statistically distinguishable from zero.

Interpretation of Heterogeneity:

The pattern of significant effects aligns with expectations of healthcare demand and access:

Groups with higher baseline need for care (e.g., those in poor health) or lower barriers to utilization (e.g., White individuals) show stronger responses to insurance.

The lack of significance for other groups suggests that while insurance eligibility removes financial barriers, it may not be sufficient to increase utilization in the absence of other enabling factors or preferences for care.

```
###2
```{r }
ohie_data <- read.csv("/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw3/OHIE.csv")
ohie_data$age_group <- ifelse(ohie_data$age < 50, 0, 1)
# 验证age_group是否创建成功
cat("age_group取值分布：\n")
print(table(ohie_data$age_group, useNA = "ifany"))  # 会显示0/1的样本量，确认无空

# 2. 定义8个分组
groups <- list(
  "Female (female=1)" = ohie_data[ohie_data$female == 1, ],
  "Male (female=0)" = ohie_data[ohie_data$female == 0, ],
  "Age ≥ 50 (age_group=1)" = ohie_data[ohie_data$age_group == 1, ],
  "Age < 50 (age_group=0)" = ohie_data[ohie_data$age_group == 0, ],
  "White (race_white=1)" = ohie_data[ohie_data$race_white == 1, ],
  "Non-White (race_white=0)" = ohie_data[ohie_data$race_white == 0, ],
  "Good Baseline Health (health_baseline=1)" = ohie_data[ohie_data$health_baseline == 1, ],
  "Poor Baseline Health (health_baseline=0)" = ohie_data[ohie_data$health_baseline == 0, ]
)

# 3. 创建结果数据框
group_result <- data.frame(
  Group = names(groups),
  beta1_treated = NA,     # treated系数
  SE_beta1 = NA,          # 标准误
  t_stat = NA,            # t统计量
  p_value = NA,           # p值（双侧）
  Sample_Size = NA        # 新增：分组样本量，方便排查
)
```
```

```r
# 4. 循环分群回归
for (i in 1:length(groups)) {
  # 当前分组的数据集和名称
  current_df <- groups[[i]]
  current_name <- names(groups)[i]
  # 记录分组样本量
  group_result$Sample_Size[i] <- nrow(current_df)

  # 跳过空分组
  if (nrow(current_df) == 0) {
    cat("\n跳过空分组: ", current_name, "\n")
    next
  }
  # 分群回归
  reg <- lm(count_visit_dr ~ treated + numhh_list, data = current_df)
  # 从回归摘要中提取treated的结果
  # summary(reg)$coefficients 会返回系数表, 直接取"treated"行
  coef_info <- summary(reg)$coefficients["treated", ]
  group_result$beta1_treated[i] <- round(coef_info["Estimate"], 4)  # 系数
  group_result$SE_beta1[i] <- round(coef_info["Std. Error"], 4)     # 标准误
  group_result$t_stat[i] <- round(coef_info["t value"], 4)          # t统计量
  group_result$p_value[i] <- round(coef_info["Pr(>|t|)"], 4)        # p值
}

# 5. 输出最终结果
cat("Question 2: Final Grouped Regression Results\n")
print(group_result)
```

```r
# 6. 标注最大/最小处理效应群体
valid_rows <- !is.na(group_result$beta1_treated)
if (sum(valid_rows) > 0) {
  max_idx <- which.max(group_result$beta1_treated[valid_rows])
  min_idx <- which.min(group_result$beta1_treated[valid_rows])
  cat("\nGroup with Max Treatment Effect:", group_result$Group[valid_rows][max_idx],
      ", Coefficient =", group_result$beta1_treated[valid_rows][max_idx], "\n")
  cat("Group with Min Treatment Effect:", group_result$Group[valid_rows][min_idx],
      ", Coefficient =", group_result$beta1_treated[valid_rows][min_idx], "\n")
}
cat("Significance Rule: p_value < 0.05 = significant at 5% level; otherwise non-significant\n")

```
```
```

3.

```
========= Question 3: Education Variable Regression Results =========
                                          beta1.treated SE     R2 Adj_R2
Baseline (numhh_list only)                       0.5935 NA 0.0063 0.0061
Model 3 (+ hs_degree + college_degree)           0.5745 NA 0.0096 0.0093
Model 4 (+ NO_hs_degree + hs + college)              NA NA     NA     NA
```

Model 3 vs. Baseline:

The treated coefficient decreases slightly from **0.5935 to 0.5745**, a change of **-0.0190** . This is a small change, confirming that education variables do not introduce substantial bias, which aligns with the randomization of the RCT.

The R^2 increases from **0.0063 to 0.0096**, and adjusted R^2 from **0.0061 to 0.0093**, indicating that education variables improve the model's explanatory power.

The standard errors (SE) are reported as NA in the current output, which is due to an issue in how SE is extracted. After fixing the get_metrics function (as discussed earlier), we expect the SE to decrease, reflecting improved estimation precision.

Model 4 Unestimable:

Model 4 cannot be estimated due to **perfect multicollinearity (the dummy variable trap)**. The variables NO_hs_degree, hs_degree, and college_degree are mutually exclusive and exhaustive (every observation belongs to exactly one category), creating a perfect linear relationship:

NO_hs_degree+hs_degree+college_degree=1

This violates the OLS assumption of no perfect multicollinearity, making it impossible to estimate unique coefficients.

```
###3
```{r }
ohie_data <- read.csv("/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw3/OHIE.csv")
# 基准模型：仅控制treated + numhh_list（与Question 1的多元回归一致）
reg_base <- lm(count_visit_dr ~ treated + numhh_list, data = ohie_data)
# Model 3：控制treated + numhh_list + hs_degree + college_degree（无虚拟变量陷阱）
reg3 <- lm(count_visit_dr ~ treated + numhh_list + hs_degree + college_degree, data = ohie_data)
# Model 4：控制treated + numhh_list + NO_hs_degree + hs_degree + college_degree（有陷阱，尝试运行）
reg4 <- try(lm(count_visit_dr ~ treated + numhh_list + NO_hs_degree + hs_degree + college_degree, data = ohie_data), silent =
TRUE)

# 定义函数：提取回归的核心指标（treated系数、标准误、R²、调整后R²）
get_metrics <- function(reg) {
  if (inherits(reg, "lm")) {  # 判断是否为有效回归结果（避免Model 4报错）
    beta1 <- coef(reg)["treated"]  # treated系数
    se <- sqrt(vcov(reg))["treated"]  # 标准误
    r2 <- summary(reg)$r.squared  # R²（拟合度）
    adj_r2 <- summary(reg)$adj.r.squared  # 调整后R²（消除变量数影响）
    return(round(c(beta1 = beta1, SE = se, R2 = r2, Adj_R2 = adj_r2), 4))  # 四舍五入
  } else {
    return(c(beta1 = NA, SE = NA, R2 = NA, Adj_R2 = NA))  # Model 4无效时返回NA
  }
}

# 提取三个模型的指标并对比
edu_result <- rbind(
  "Baseline (numhh_list only)" = get_metrics(reg_base),  # 基准模型
  "Model 3 (+ hs_degree + college_degree)" = get_metrics(reg3),  # 无陷阱模型
  "Model 4 (+ NO_hs_degree + hs + college)" = get_metrics(reg4)   # 有陷阱模型
)
```

4.

```
========= Question 4: Full Control Model Comparison =========
                          beta1.treated SE     R2 Adj_R2
Baseline (numhh_list only)        0.5935 NA 0.0063 0.0061
Full Control Model                0.5856 NA 0.0326 0.0319
```

Unbiasedness: The treated coefficient changes minimally from 0.5935 to 0.5856 , which is well below 0.05. This confirms that randomization ensures baseline variables are independent of treated, so omitting them does not introduce substantial bias.

Estimation Precision: The standard errors (SE) are reported as NA in the current output, which indicates an issue with the regression output extraction. After fixing the get_metrics function to correctly extract SE from summary(reg)$coefficients, we expect the SE to decrease, reflecting improved precision due to reduced residual variance.
```

Model Fit: The R-squared increases dramatically from 0.0063 to 0.0326, and the adjusted R-squared from 0.0061 to 0.0319. This represents a more than 400% improvement in model fit, indicating that baseline variables have strong explanatory power for variation in doctor visits.

```{r }
ohie_data <- read.csv("/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw3/OHIE.csv")
reg_full <- lm(
  count_visit_dr ~ treated + numhh_list + female + age + race_white +
    hs_degree + college_degree + health_baseline,
  data = ohie_data
)

# Compare baseline model and full control model
full_compare <- rbind(
  "Baseline (numhh_list only)" = get_metrics(reg_base),
  "Full Control Model" = get_metrics(reg_full)
)
```

5.

Only controlling 'numhh_list' ensures the 'treated' (insurance effect) is unbiased; full baseline controls are unnecessary.

Reason: OHIE is a Randomized Controlled Trial (RCT). 'treated' is randomly assigned and pre-independent of other baseline variables (gender, age, race, etc.). Omitting these variables does not cause bias. Only 'numhh_list' disrupts random balance due to the study design (insurance assigned at the household level), so it needs to be controlled.

6.

Full baseline controls should be included to improve the estimation precision of the 'treated' (insurance effect).

Reason: These baseline variables are highly correlated with the outcome (doctor visits) but independent of 'treated' (ensured by randomization). Including them reduces residual variance (fluctuations in regression error), which lowers the SE of the 'treated' coefficient and improves precision. With a large sample size, the cost of lost degrees of freedom is negligible, and the benefit (precision gain) far outweighs the cost.

7.

	theta_hat <dbl>	se_theta <dbl>	t_stat <dbl>	df <int>	p_value <dbl>	Conclusion <chr>
treated	-0.3336	0.351	-0.9503	12116	0.342	Fail to Reject H0 (No significant difference)

At the 5% significance level, we **fail to reject the null hypothesis** (H0). The p-value of 0.342 is much larger than 0.05, indicating that there is no statistically significant difference between the effect of health insurance eligibility and the effect of having a high school diploma on doctor visits.

This result is different from the typical OHIE findings and suggests that, in this specific sample, health insurance and a high school education have statistically indistinguishable impacts on healthcare utilization.

```
###7
```{r }
ohie_data <- read.csv("/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw3/OHIE.csv")
# H0: beta_treated = beta_hs_degree (Insurance effect = High school degree effect)
##################################################
cat("\n========= Question 7: Manual Hypothesis Test =========\n")
# Extract coefficients and variance-covariance matrix from full control model (Question 4)
beta <- coef(reg_full)  # All coefficients (including 'treated' and 'hs_degree')
vcov_mat <- vcov(reg_full)  # Variance-covariance matrix (base R function, no extra packages)

# Extract core parameters for testing
beta_t <- beta["treated"]       # Coefficient for 'treated' (insurance effect)
beta_hs <- beta["hs_degree"]    # Coefficient for 'hs_degree' (high school effect)
var_t <- vcov_mat["treated", "treated"]  # Variance of 'treated' coefficient
var_hs <- vcov_mat["hs_degree", "hs_degree"]  # Variance of 'hs_degree' coefficient
cov_th <- vcov_mat["treated", "hs_degree"]   # Covariance between 'treated' and 'hs_degree' coefficients

# Manually calculate test statistics (H0: beta_t = beta_hs → theta = beta_t - beta_hs = 0)
theta_hat <- beta_t - beta_hs  # Estimated theta (difference between coefficients)
se_theta <- sqrt(var_t + var_hs - 2 * cov_th)  # SE of theta (variance formula: Var(a-b) = Var(a)+Var(b)-2Cov(a,b))
t_stat <- theta_hat / se_theta  # t-statistic (tests if theta is significantly different from 0)
df_res <- reg_full$df.residual  # Residual degrees of freedom (for p-value calculation)
p_val <- 2 * pt(abs(t_stat), df = df_res, lower.tail = FALSE)  # Two-tailed p-value
```

```
# Organize test results (round numeric values)
test_result <- data.frame(
  theta_hat = round(unname(theta_hat), 4),  # Estimated theta
  se_theta = round(unname(se_theta), 4),    # SE of theta
  t_stat = round(unname(t_stat), 4),        # t-statistic
  df = df_res,                              # Degrees of freedom
  p_value = round(unname(p_val), 4),        # p-value
  Conclusion = ifelse(p_val < 0.05, "Reject H0 (Insurance effect ≠ High school effect)", "Fail to Reject H0 (No significant
difference)")
)
```
```